



SUPPLY CHAIN INTELLIGENCE // FREE CASE STUDY // SPACE DEFENSE // US

TURION SPACE

Golden Dome Prime Contractor Running on a Propellant It Cannot Guarantee

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Turion Space is one of twelve prime contractors selected for the US Space Force Golden Dome Space-Based Interceptor programme — alongside Lockheed Martin, Northrop Grumman, SpaceX, and Anduril. Its DROID spacecraft performs rendezvous, proximity operations, and inspection. Its propulsion system depends on xenon — a noble gas produced by exactly five facilities globally. Two are in China. The US strategic xenon reserve is classified. And Turion has no disclosed long-term xenon supply agreement. This case study maps what happens when a spacecraft propellant becomes a national security chokepoint.

5	<10%	\$7,200	DROID
GLOBAL XENON	US SHARE OF GLOBAL	XENON PRICE PER KG	DROID SPACECRAFT
PRODUCTION FACILITIES — 2 IN CHINA	XENON PRODUCTION CAPACITY	UP 340% SINCE 2020	RPO + INSPECTION — HALL THRUSTER

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defense.codes | ceo@capa.cloud | +91 91637 62717

INTELLIGENCE ALERT // TURION SPACE // THE XENON CHOKEPOINT

THE CORE FINDING

Turion Space's DROID spacecraft uses Hall-effect thrusters for orbital manoeuvring. Hall-effect thrusters run on xenon. Xenon is produced as a byproduct of oxygen separation at steel mills and air separation plants. Five facilities globally dominate production. Two are in China. The US produces less than 10% of global supply. Xenon prices have increased 340% since 2020. Turion has no disclosed long-term xenon supply agreement. For a company building spacecraft that must manoeuvre to intercept missiles at GEO altitude, running out of propellant is not a logistics problem. It is a mission failure.

THE XENON SUPPLY PROBLEM // WHY SPACE PROPELLANT IS A NATIONAL SECURITY ISSUE

Xenon is extracted from air as a byproduct of industrial oxygen production — a process dominated by large air separation units at steel mills and chemical plants. It cannot be produced on demand; it is recovered during oxygen separation and stored. Global annual production is approximately 40 million litres — enough for approximately 60,000 kg of liquid xenon. The Hall-effect thruster market for space applications consumes roughly 8,000–12,000 kg annually. As LEO constellations proliferate and space defence missions multiply, xenon demand is growing 25–35% annually. Supply is not.

RISK FACTOR	DETAIL	TURION SPECIFIC IMPACT	SEVERITY
Geographic concentration	5 primary global facilities: 2 PRC, 1 Russia, 1 Germany (Linde), 1 US (Air Products)	60%+ of global xenon supply in adversary or at-risk geography	CRITICAL
US domestic production	<10% of global xenon capacity. Air Products Allentown is the primary US facility — single plant	US xenon reserve insufficient for scaled Golden Dome propulsion without import	CRITICAL
Price trajectory	Xenon \$2,100/kg in 2020 — \$7,200/kg Q2 2026 — +340%. Driven by LEO constellation demand	Each DROID spacecraft carrying 50–100 kg propellant: \$360K–\$720K propellant cost per spacecraft	HIGH
No Turion supply agreement	No long-term xenon supply contract identified in any public filing or press release	Spot market exposure — price and availability risk unmitigated	CRITICAL
Alternative propellants	Krypton cheaper but lower performance. AF-M315E green monoprop — no Hall thruster. Electric alternatives years away	Krypton is a partial option — approximately 20% performance penalty	HIGH

THE GOLDEN DOME CONTEXT // WHY PROPELLANT SUPPLY IS A MISSION ASSURANCE ISSUE

The Golden Dome Space-Based Interceptor mission requires spacecraft that can manoeuvre to intercept ballistic and hypersonic missiles at altitude. The delta-V budget for intercept operations — the propellant required to change orbit, intercept a trajectory, and return to patrol orbit — is substantially higher than the depot-maintenance RPO missions that most commercial space tug operators plan for. At \$7,200/kg xenon and 50–100 kg per DROID spacecraft, a constellation of 50 DROIDS requires \$18M–\$36M in xenon before refuelling. If PRC export controls on xenon are applied — a step China has not yet taken but has the mechanism to implement — US xenon availability for defence missions becomes a direct function of Air Products' Allentown throughput.

The supply chain vulnerability for Turion Space is not about whether DROID works. The RPO capability is real. The Golden Dome selection is real. The vulnerability is whether the propellant that makes DROID manoeuvre will be available at the price and quantity required when Golden Dome moves from prototype to constellation. That question has no publicly available answer. Which means the answer is: nobody has secured it yet.

WHAT A FULL ENGAGEMENT WOULD ADD

A full Defense.Codes engagement would add: (1) Xenon supply chain map — all five global facilities, annual throughput, contract calendar, and US DOE strategic reserve status; (2) Turion propellant architecture analysis from patent and technical paper review; (3) Krypton transition cost-benefit model; (4) Golden Dome propellant logistics scenario planning; (5) Competitor propellant strategy comparison — True Anomaly, Turion, Impulse Space.

This case study is one of five free public intelligence samples published by Defense.Codes. Category: SPACE DEFENSE — US. Full-depth engagement reports available for purchase. Contact: ceo@capa.cloud | defense.codes

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Sources: Space Force Golden Dome selection announcement Apr 2026 — The Next Web Apr 2026 — Washington Technology Apr 2026 — Turion Space website and press releases — Air Products xenon production disclosures — Fastmarkets xenon pricing Q2 2026 — CSIS critical minerals analysis 2025 — ESA propellant demand analysis 2024.

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